



# Wi-Fi Security Protocols

## ◆ Introduction

Wi-Fi security protocols are used to **protect wireless networks** from unauthorized access, data theft, and attacks. They provide **authentication, encryption, and data integrity**.

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## ◆ 1. WEP (Wired Equivalent Privacy)

### Overview:

- First Wi-Fi security protocol (IEEE 802.11)
- Uses **RC4 encryption algorithm**
- Key sizes: **64-bit / 128-bit**

### Features:

- Static key (same key for all users)
- Uses Initialization Vector (IV)

### Advantages:

- Simple to implement

### Disadvantages:

- Very weak security
- Easily hackable due to **IV reuse and weak encryption**

### Status:

 **Obsolete (not recommended)**

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## ◆ 2. WPA (Wi-Fi Protected Access)

### Overview:

- Introduced as an improvement over WEP
- Uses **TKIP (Temporal Key Integrity Protocol)**

#### **Features:**

- Dynamic key generation
- Message Integrity Check (MIC)

#### **Advantages:**

- More secure than WEP
- Backward compatible

#### **Disadvantages:**

- Still vulnerable to attacks
- Uses RC4 (outdated)

#### **Status:**

#### **Deprecated**

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## ◆ **3. WPA2 (Wi-Fi Protected Access 2)**

#### **Overview:**

- Based on **IEEE 802.11i standard**
- Uses **AES encryption (Advanced Encryption Standard)**

#### **Features:**

- Strong encryption
- Two modes:
  - **Personal (PSK)** → Password-based
  - **Enterprise** → Uses authentication server (RADIUS)

#### **Advantages:**

- Highly secure
- Widely used

#### **Disadvantages:**

- Vulnerable to:

- Weak passwords
- KRACK attack

 **Status:**

 **Still widely used**

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## ◆ 4. WPA3 (Wi-Fi Protected Access 3)

 **Overview:**

- Latest Wi-Fi security protocol
- Introduced to overcome WPA2 weaknesses

 **Features:**

- Uses **SAE (Simultaneous Authentication of Equals)**
- Stronger encryption and protection

 **Advantages:**

- Better protection against password guessing
- Forward secrecy
- Strong encryption even with weak passwords

 **Disadvantages:**

- Requires modern hardware
- Not supported on older devices

 **Status:**

 **Recommended (latest standard)**

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## ◆ Comparison Table

| Protocol | Encryption | Security Level | Status   |
|----------|------------|----------------|----------|
| WEP      | RC4        | Very Low       | Obsolete |

|      |            |           |             |
|------|------------|-----------|-------------|
| WPA  | RC4 + TKIP | Medium    | Deprecated  |
| WPA2 | AES        | High      | Widely Used |
| WPA3 | SAE + AES  | Very High | Latest      |

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## ◆ Key Concepts

### Authentication

- Verifies user identity
- Example: Password, RADIUS server

### Encryption

- Protects data from attackers
- Example: AES

### Integrity

- Ensures data is not altered
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